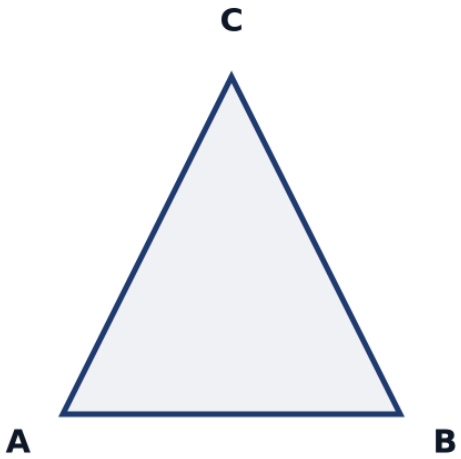
Advanced Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

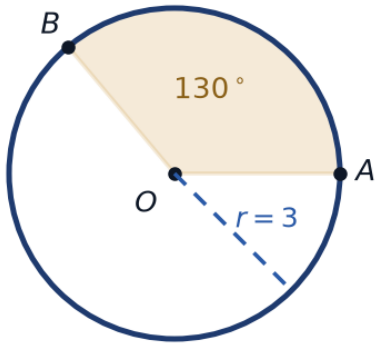
1 [Foundational] • **Target: Distance, midpoint, and slope combined**

Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB.



2 [Foundational] • **Target: Equations of circles from given conditions**

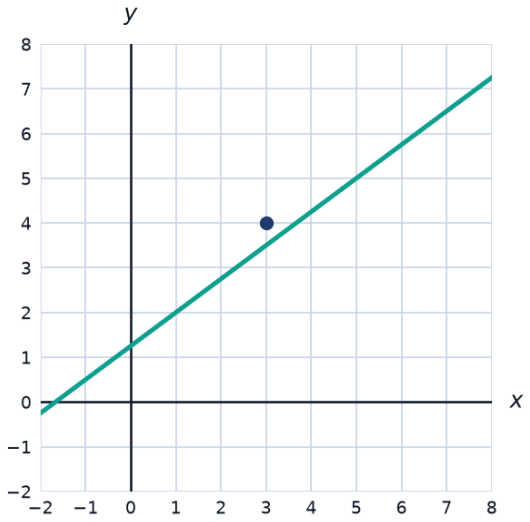
Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.



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3 [Foundational] • Target: Distance from a point to a line

Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.



4 [Foundational] • Target: Locus problems

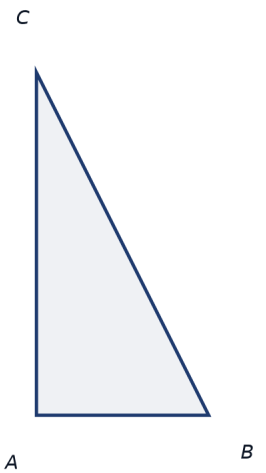
Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.

5 [Foundational] • Target: Rotations and general transformations

A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \to (-y, x)$.

6 [Foundational] • Target: Dilations and scale factor

A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.



7 [Foundational] • Target: Finding the equation of a parabola from three points

A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a, b, and c.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: A multi-part triangle-in-the-plane problem**

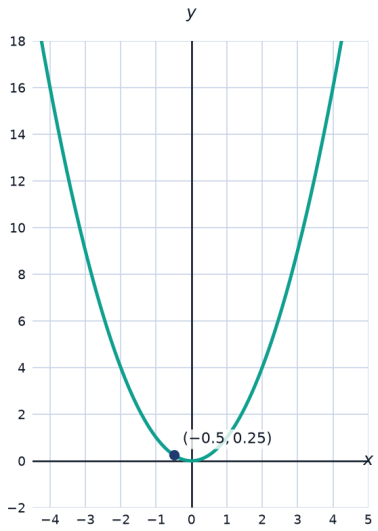
This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula Area $= \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.

9 **[Intermediate]** • **Target: Finding the equation of a line given a point and a perpendicular condition**

Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.

10 **[Intermediate]** • **Target: Translating and reflecting equations of curves**

The graph of $y = x^2$ is reflected over the x-axis, then shifted 3 units up. Write the equation of the resulting graph.



11 **[Intermediate]** • **Target: Finding vertices of a polygon given transformations**

A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.

12 **[Intermediate]** • **Target: Finding the intersection of a line and a parabola**

Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.

13 **[Intermediate]** • **Target: A multi-part quadrilateral classification problem**

This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 0)$, $Y(8, 4)$, $Z(2, 4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY, and of WZ and XY. (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.

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14 [Intermediate] • Target: Finding the equation of a tangent line to a circle

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.

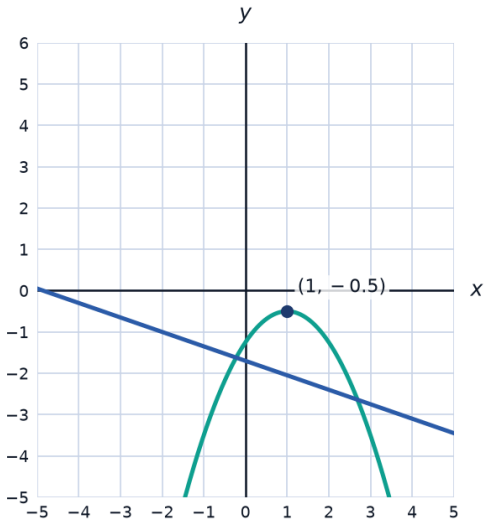
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding the vertex and axis of symmetry of a parabola from three points • Cross-Lesson: 7.01 Advanced Plane Geometry

A parabola passes through $(-1, 0)$, $(3, 0)$, and $(1, -8)$. Find the axis of symmetry and the vertex.

16 [Challenging] • Target: Distance, midpoint, and slope combined • Cross-Lesson: 8.01 Statistics, Counting & Probability

Find the standard equation $Ax + By = C$ of the line perpendicular to $3x - 4y = 12$ that passes through the midpoint of the line segment joining $P(-2, 7)$ and $Q(6, 1)$.



17 [Challenging] • Target: Equations of circles from given conditions • Cross-Lesson: 1.01 Advanced Number Theory

Find the shortest perpendicular distance from the point $P(2, -3)$ to the line $3x - 4y + 7 = 0$.

18 [Challenging] • Target: Distance from a point to a line • Cross-Lesson: 2.01 Advanced Algebra & Systems

Find the distance between points $A(1, 2, 3)$ and $B(4, 6, 3)$, and find the midpoint M of the line segment AB.

19 [Challenging] • Target: Locus problems • Cross-Lesson: 3.01 Advanced Functions

An ellipse has the equation $9x^2 + 25y^2 = 225$. Determine its semi-major axis a, semi-minor axis b, the coordinates of its foci, and its eccentricity e.

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20 [Challenging] • Target: Rotations and general transformations • Cross-Lesson: 4.01 Advanced Trigonometry

A quadrilateral $ABCD$ is inscribed in a circle. If $\angle DAB = (2x + 15)^\circ$ and the opposite angle $\angle DCB = (3x - 10)^\circ$, determine the value of x and calculate the measure of $\angle DAB$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Distance, midpoint, and slope combined

Find the slope of the line through $(4, -5)$ and $(8, -2)$.

a. $-\frac{3}{4}$

b. $\frac{3}{4}$

c. $-\frac{4}{3}$

d. $\frac{4}{3}$

22 [Foundational] • Target: Equations of circles from given conditions

Find the y-intercept of the perpendicular bisector of the segment connecting $(-2, -1)$ and $(4, -6)$.

a. $-\frac{47}{10}$

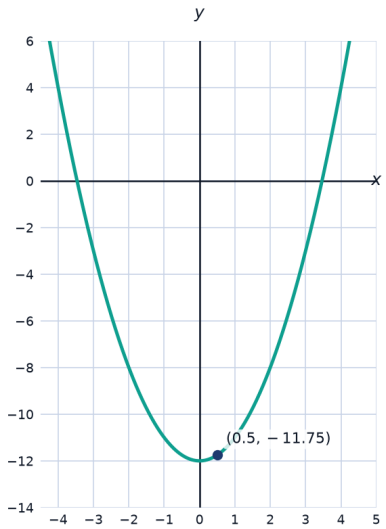
b. $-\frac{8}{3}$

c. $-\frac{37}{10}$

d. $\frac{47}{10}$

23 [Foundational] • Target: Distance from a point to a line

Find the sum of the x-coordinates of the intersection points of $y = x^2 - 12$ and $y = -4x$.



a. -12

b. -4

c. -2

d. -6

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24 [Foundational] • Target: Locus problems

In the standard (x, y) coordinate plane, the line $y = mx$ (where $m > 0$) is tangent to the circle $(x - 10)^2 + y^2 = 36$. What is the value of m?

a. $\frac{3}{5}$

b. $\frac{3}{4}$

c. $\frac{4}{5}$

d. $\frac{4}{3}$

25 [Foundational] • Target: Rotations and general transformations

In the standard (x, y) coordinate plane, the vertices of ΔOAB are $O(0, 0)$, $A(6, 0)$, and $B(2, 4)$. A vertical line with equation $x = k$, where $0 < k < 6$, divides ΔOAB into two regions of equal area. What is the value of k?

a. $6 - 2\sqrt{3}$

b. $\sqrt{6}$

c. $\frac{8}{3}$

d. 3

26 [Foundational] • Target: Dilations and scale factor

In the standard (x, y) coordinate plane, line L is perpendicular to the line $3x + 4y = 15$ and passes through the point $(3, 1)$. Line L intersects the line $2x + y = 17$ at the point (p, q). What is the value of $p + q$?

a. 5

b. 6

c. 8

d. 11

27 [Foundational] • Target: Finding the equation of a parabola from three points

In the figure below, ΔABC is a right triangle with hypotenuse $\overline{AB}$, and altitude $\overline{CD}$ of length h is drawn to $\overline{AB}$. The altitude divides the hypotenuse into two segments of lengths $AD = 9$ and $DB = 16$. A circle with center O_1 is inscribed in ΔACD, and a circle with center O_2 is inscribed in ΔBCD. What is the distance between the centers O_1 and O_2?

a. 5

b. $5\sqrt{2}$

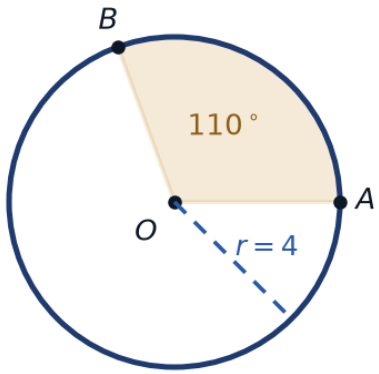
c. 7

d. $7\sqrt{2}$

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28 [Foundational] • Target: A multi-part triangle-in-the-plane problem

In the standard (x, y) coordinate plane, what are the center and the radius of the circle with equation $(x-4)^2+(y+6)^2=49$?



a. Center $(4, -6)$, radius 7

b. Center $(-4, 6)$, radius 7

c. Center $(4, -6)$, radius 49

d. Center $(-4, 6)$, radius 49

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding the equation of a line given a point and a perpendicular condition

In the standard (x, y) coordinate plane, a circle has its center at $(-3, 5)$ and passes through the point $(1, 8)$. What is the radius of this circle?

a. 5

b. 7

c. $\sqrt{13}$

d. 25

30 [Intermediate] • Target: Translating and reflecting equations of curves

In the standard (x, y) coordinate plane, what is the length of the major axis of the ellipse defined by the equation $\frac{(x-4)^2}{49}+\frac{(y+2)^2}{16}=1$?

a. 7

b. 8

c. 14

d. 49

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31 [Intermediate] • Target: Finding vertices of a polygon given transformations

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. 25

b. 50

c. 100

d. 125

32 [Intermediate] • Target: Finding the intersection of a line and a parabola

In the xy-plane, let $P(t, t^2)$ with $t > 0$ be a point on the parabola $y = x^2$. The normal line to the parabola at P intersects the parabola again at a second point Q. Let T be the intersection point of the tangent lines to the parabola at P and Q. As $t \to 0^+$, what is the limiting value of the y-coordinate of T?

a. $-\frac{1}{2}$

b. $-\frac{1}{4}$

c. -1

d. 0

33 [Intermediate] • Target: A multi-part quadrilateral classification problem

A circle has center $(4, -2)$ and passes through $(13, 10)$. Find the slope of the tangent line to the circle at $(13, 10)$.

a. 5/4

b. 1/4

c. -3/4

d. -11/4

34 [Intermediate] • Target: Finding the equation of a tangent line to a circle

In ΔABC, the side lengths are $AB = 9$, $AC = 8$, and $BC = 7$. A line segment $\overline{DE}$, with D lying on $\overline{AB}$ and E lying on $\overline{AC}$, divides ΔABC into two regions that have equal areas and equal perimeters. What is the length of $\overline{DE}$?

a. $2\sqrt{6}$

b. $\frac{7\sqrt{2}}{2}$

c. $4\sqrt{3}$

d. $6\sqrt{2}$

35 [Intermediate] • Target: Finding the vertex and axis of symmetry of a parabola from three points

Each interior angle of a regular polygon measures 144°. How many sides does the polygon have?

a. 8

b. 9

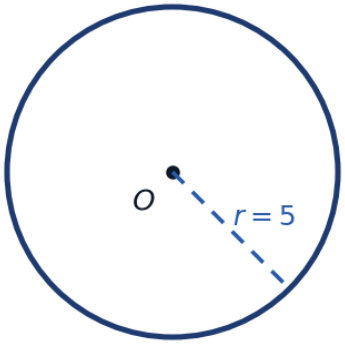
c. 10

d. 12

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36 [Intermediate] • Target: Distance, midpoint, and slope combined

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.



a. 25

b. 30

c. 35

d. 40

37 [Intermediate] • Target: Equations of circles from given conditions

In right triangle ABC, altitude $\overline{BD}$ is drawn to hypotenuse $\overline{AC}$. If $AD = 4$ and $DC = 9$, what is the length of altitude $\overline{BD}$?

a. 6

b. 6.5

c. 13

d. 36

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Distance from a point to a line • Cross-Lesson: 2.01 Advanced Algebra & Systems

Find the radius of the circle $x^2 + y^2 + 4x + 12y + 15 = 0$.

a. 5

b. 25

c. 6

d. 4

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39 [Challenging] • Target: Locus problems • Cross-Lesson: 3.01 Advanced Functions

Find the distance between $(-2, -1)$ and $(-6, -4)$.

a. 5

b. 6

c. 4

d. 7

40 [Challenging] • Target: Rotations and general transformations • Cross-Lesson: 4.01 Advanced Trigonometry

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. 4

c. 5

d. 6

41 [Challenging] • Target: Dilations and scale factor • Cross-Lesson: 5.01 Pre-Calculus Concepts

A circle $\mathcal{C}$ with center $C(0, k)$ on the positive y-axis passes through the focus F of the parabola $x^2 = 4y$. If $\mathcal{C}$ is tangent to the parabola at two distinct points, P and Q, what is the area of quadrilateral $PFQC$?

a. $8\sqrt{3}$

b. $4\sqrt{3}$

c. $16\sqrt{3}$

d. $6\sqrt{3}$

42 [Challenging] • Target: Finding the equation of a parabola from three points • Cross-Lesson: 7.01 Advanced Plane Geometry

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. 1

d. 5/3

43 [Challenging] • Target: A multi-part triangle-in-the-plane problem • Cross-Lesson: 8.01 Statistics, Counting & Probability

A circle has center $(6, 6)$ and radius 2. Find the shortest distance from the point $(16, 30)$ to the circle.

a. 6

b. 38

c. 24

d. 29

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44 [Challenging] • Target: Finding the equation of a line given a point and a perpendicular condition • Cross-Lesson: 1.01 Advanced Number Theory

Line ℓ_1 has slope -2. What is the slope of a line perpendicular to ℓ_1?

a. 1/2

b. 5/2

c. 1/4

d. 1/6

45 [Challenging] • Target: Translating and reflecting equations of curves • Cross-Lesson: 2.01 Advanced Algebra & Systems

The line $y = -12$ intersects the circle $x^2 + y^2 = 169$ at two points. What is the distance between the two intersection points?

a. 10

b. 13

c. 16

d. 5

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